Knowledge Organiser
Year 8: Restless Earth – The Solar System

The Planets of the Solar System:
- There are 8 planets in the solar system: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune
- We can use mnemonics (memory devices) to help us remember the order



A mnemonic used to remember the order of the planets in the solar system

What about Pluto? What makes a planet?
- In 2006, the International Astronomical Union (IAU) redefined the criteria for planet classification.
- A planet must:

1. Orbit the Sun
2. Be spherical in shape
3. Clear its orbit of other debris

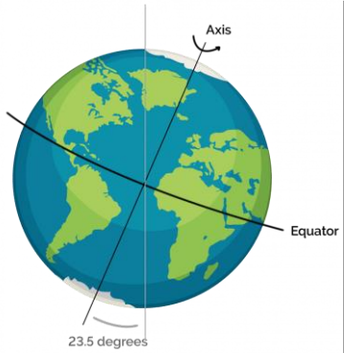
- Pluto does not clear its orbit and is now classified as a "dwarf planet."



Orbits: The Earth and the Moon
- The moon orbits the Earth approximately every 27.3 days
- This is called a "lunar month"
- The Earth orbits the Sun in an oval shape
- One complete orbits takes 365.25 days
- We call this a "solar year"

Event	Time taken
Lunar month	27.3 days
Calendar month	30, 31 or 28 days
Solar year	365.25 days
Calendar year	365 days (+1 for leap years)
Earth rotates on axis	24 hours

A table which gives the time taken for various events in the solar system



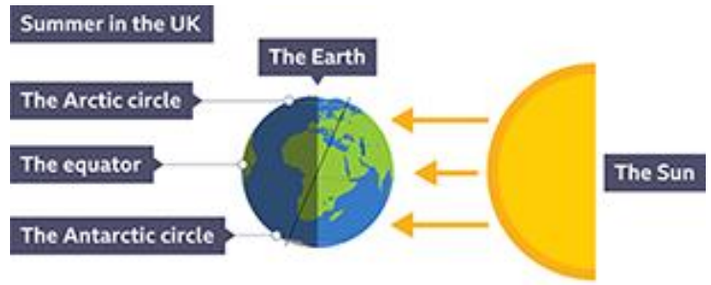
Earth's Axis of Rotation:
- The Earth rotates around an imaginary "axis of rotation"
- This axis is tilted at an angle of 23.5 degrees, it doesn't go through the "top" or "bottom" of the Earth

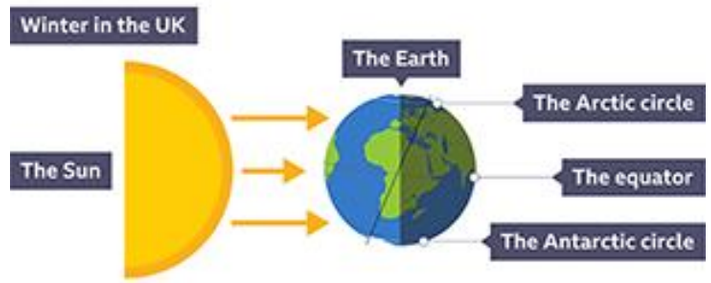


A diagram which explains why we experience a day and night cycle

Days and nights:
- The Earth takes around 24 hours to rotate fully
- At some points in this rotation each country is facing the Sun directly, this is day time
- At other points each country is facing away from the Sun, this is the night time

Why do we have seasons?
- The tilt of the Earth's axis of rotation causes different parts of Earth to receive varying amounts of Sunlight throughout the year
- The two "halves" of the world are called the Northern and Southern hemispheres
- During Summer in the Northern hemisphere it is Winter in the Southern hemisphere
- This is because only one hemisphere can be tilted towards the Sun at a time





A diagram which explains why the UK experiences different weather during the Summer and Winter months

RAG Yourself!

Learning Outcomes;	RAG
State the order of the planets in the Solar System	
Explain why Pluto is no longer regarded as a planet	
Describe the Earth's orbit around the Sun	
State the length of a day, lunar month, and solar year	
Explain how the tilt of the Earth is linked to seasons	
Explain why it is Summer in the Northern hemisphere when it is Winter in the Southern hemisphere	

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The Planets of the Solar System:

A mnemonic used to remember the order of the planets in the solar system

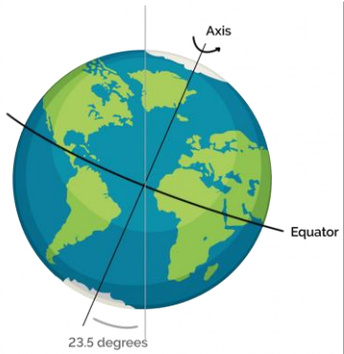
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Orbits: The Earth and the Moon

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Calendar month	
Solar year	
Calendar year	
Earth rotates on axis	

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Earth's Axis of Rotation:

A diagram which explains why we experience a day and night cycle

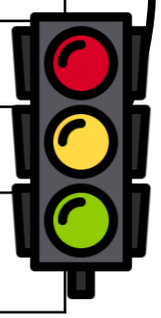
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